

	$V = \sqrt{(gr)} = \sqrt{(5 \times 9.81)} = 3.4 \text{ m/s}$ weight
9	Ans : A Gravitational field strength, $g = \frac{GM}{r^2}$ Since $M = \rho \times V = \rho \times \frac{4}{3}\pi r^3$, $g = \frac{\left(G \times \rho \times \frac{4}{3}\pi r^3\right)}{r^2} = \frac{4}{3}G\rho\pi r$, i.e. g is proportional to r. Therefore, $\frac{g_x}{g_r} = \frac{R}{2R} = 0.5$
10	Ans : B